

16.4) Find the magnitude and direction of the force on the charge $-Q$ in Fig. 16.35.

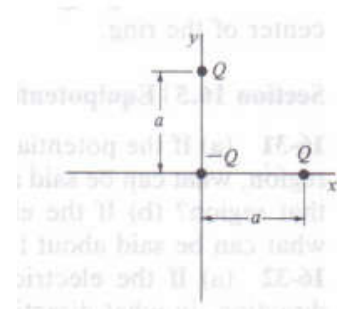


Figure 16.35. Exercise 16-4.

16.5) In a NaCl molecule, a Na^+ ion with charge e is 2.3×10^{-10} m from a Cl^- ion with charge $-e$. What is the magnitude of the force between them?

16.11) An electron is accelerated at 10^8 m s^{-2} by an electric field. What is the direction and magnitude of the field?

16.21) A carbon nucleus has a charge of $+6e$. At a distance of 10^{-10} m from a carbon nucleus, find (a) the electric potential; (b) the potential energy of an electron in electron volts and joules.

16.23) Two uniformly charged metal plates separated by 0.04 m produce a uniform field between them of 10^4 N C^{-1} . Find (a) the charge per unit area Q/A on the plates; (b) the potential difference between the plates.

16.42) Two square metal plates with sides of length 0.1 m are separated in vacuum by 10^{-3} m. Find their capacitance.

16.54) A $500\mu\text{F}$ capacitor is connected to 100 V potential difference. How much work is stored in the capacitor?

Q(1) Find the electric force between a proton and the electron in a Hydrogen atom.

Q(2) A charge $+q = 8\mu\text{C}$ of mass 0.005 kg subject to external electric field $E = 1500 \text{ V/m}$. What is the acceleration of the charge?

Q(3) What is the meaning of a uniform electric field E ?

Q(4) What is the electrostatic potential energy U stored in a configuration of two electric charges if they are separated by 3m assuming $q_1 = +2\mu\text{C}$ and $q_2 = -5\mu\text{C}$?